SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 1 **Duration : 1 Hour**
Total Marks:10

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I RAHIMA BANU N-192525009 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. **Problem Statement**

   A hospital needs to assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) using Backtracking Search while satisfying all the given constraints.

2. **Constraints**

1. D1 cannot work the Night shift.
2. D2 must work before D3.
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the Morning shift.
5. All doctors must be assigned exactly one shift.

3. **Backtracking Search Approach**

   Backtracking Search assigns a shift to one doctor at a time. After each assignment, the constraints are checked. If any constraint is violated, the algorithm rejects that assignment and tries another possibility.

4. **Intermediate Steps and Constraint Checks**

   Step 1: Assign D1

- Try D1 = Morning.
- Constraint 1: D1 is not assigned to Night — Satisfied.
- The assignment is valid.

  Current assignment: D1 = Morning

  Step 2: Assign D2
- Try D2 = Afternoon.
- Constraint 2: D2 must work before D3 — To be checked after assigning D3.
- Constraint 3: Morning and Afternoon are assigned to different doctors — Satisfied.

  Current assignment: D1 = Morning, D2 = Afternoon

  Step 3: Assign D3
- Try D3 = Night.
- Constraint 4: D3 cannot work Morning — Satisfied.
- Constraint 3: Only one doctor per shift — Satisfied.
- Constraint 2: D2 works Afternoon and D3 works Night, so D2 works before D3 — Satisfied.
- Constraint 5: All three doctors have exactly one shift — Satisfied.

  All constraints are satisfied. Therefore, the search terminates successfully without backtracking.

5. Constraint Verification

Constraint	Verification	Result
D1 cannot work Night	D1 works Morning	Satisfied
D2 must work before D3	Afternoon comes before Night	Satisfied
Only one doctor per shift	Each shift has one doctor	Satisfied
D3 cannot work Morning	D3 works Night	Satisfied

6. Final Valid Schedule

Shift	Assigned Doctor
Morning	D1
Afternoon	D2
Night	D3

Final Answer:

The valid schedule is Morning – D1, Afternoon – D2, Night – D3. All constraints are satisfied, and Backtracking Search finds a valid solution.